

Qi Li

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Google Scholar citations: 1200+ $\diamond$ H-index: 14

RESEARCH PROFILE

Researcher in intelligent fault diagnosis, prognostics and health management (PHM), and trustworthy industrial AI. My work develops data-efficient, explainable, and generalizable diagnosis methods for industrial equipment under variable and unseen operating conditions. Current research integrates neuro-symbolic reasoning, signal-processing knowledge, physics-informed foundation models, and industrial vibration-signal analysis to build transferable PHM systems and open benchmarks.

APPOINTMENTS AND EDUCATION

Postdoctoral Researcher Department of Automation, Tsinghua University	Jun. 2026 – Present <i>Working with Prof. Xiao He</i>
Ph.D. in Mechanical Engineering Tsinghua University	Sept. 2022 – Jun. 2026 <i>Advisor: Prof. Zhaoye Qin</i>
Visiting Researcher Yale University, Statistics and Data Science	Apr. 2025 – Sept. 2025 <i>Host: Prof. Lu Lu</i>
M.S. in Control Theory and Control Engineering Soochow University	Sept. 2019 – Jun. 2022 <i>Advisors: Prof. Liang Chen and Prof. Changqing Shen</i>
B.S. in Electrical Engineering and Automation Soochow University	Sept. 2015 – Jun. 2019

RESEARCH FOCUS

- **Trustworthy PHM and explainable diagnosis:** neuro-symbolic and interpretable architectures for fault diagnosis; evidence chains from signal-processing knowledge, symbolic regression, and self-organized neural-symbolic nodes.
- **Industrial big data under domain shift:** cross-domain adaptation, domain generalization, and augmentation for fault diagnosis under unseen working conditions and heterogeneous industrial data.
- **Physics-informed industrial foundation models:** physics-informed embeddings, backbones, and task modules for multi-sensor, multi-model, and multi-task PHM.
- **Industrial AI agents and decision support:** multi-agent signal-processing workflows and large multimodal foundation models for industrial vibration-signal interpretation and maintenance-oriented decision support.

SELECTED AWARDS AND HONOURS

- Selected for the Shuimu Scholar Program, Tsinghua University, 2026.
- Distinguished Graduate, Tsinghua University, 2026.
- Merit Student Award, Tsinghua University, 2026.
- Academic Rising Star Award, Department of Mechanical Engineering, Tsinghua University, 2026.
- National Scholarship, 2025; National Scholarship, 2021 and 2020.
- Young Elite Scientists Sponsorship Program by CAST, Ph.D. Special Plan, 2024.
- Second Prize, Science and Technology Award, Chinese Society for Vibration Engineering, 2024.
- Jiangsu Province Excellent Master's Thesis Award, 2023; Jiangsu Province Outstanding Graduate, 2022.
- Tsinghua University Future Scholar Scholarship, 2022; Tsinghua University Scholarship for Social Practice, 2023.
- Best 3MT Presentation, The University of Western Australia, 2021.

REPRESENTATIVE PUBLICATIONS

1. **Q. Li**, X. Zhang, J. Huang, H. He, F. Zhang*, Z. Qin*, F. Chu. “VSLLaVA: a pipeline of large multimodal foundation model for industrial vibration signal analysis.” *Advanced Engineering Informatics*, 2026. SCI; CAS Q1 Top; IF: 11.5.
2. **Q. Li**, B. Chen, Q. Chen, X. Li, Z. Qin, F. Chu. “HSE: A Plug-and-Play Module for Unified Fault Diagnosis Foundation Models.” *Information Fusion*, 2025, 123: 103277. IF: 15.5.
3. **Q. Li**, L. Qin, H. Xu, Q. Lin, Z. Qin, F. Chu. “Transparent information fusion network: An explainable network for multi-source bearing fault diagnosis via self-organized neural-symbolic nodes.” *Advanced Engineering Informatics*, 2025, 65: 103156. IF: 9.9.
4. **Q. Li**, Y. Liu, S. Sun, Z. Qin, F. Chu. “Deep expert network: A unified method toward knowledge-informed fault diagnosis via fully interpretable neuro-symbolic AI.” *Journal of Manufacturing Systems*, 2024, 77: 652–661. IF: 14.2.
5. **Q. Li**, H. Li, W. Hu, S. Sun, Z. Qin, F. Chu. “Transparent Operator Network: A Fully Interpretable Network Incorporating Learnable Wavelet Operator for Intelligent Fault Diagnosis.” *IEEE Transactions on Industrial Informatics*, 2024, doi: 10.1109/TII.2024.3366993. IF: 9.9.
6. **Q. Li**, L. Chen, L. Kong, D. Wang, M. Xia, C. Shen. “Cross-domain augmentation diagnosis: An adversarial domain-augmented generalization method for fault diagnosis under unseen working conditions.” *Reliability Engineering & System Safety*, 2023, 234: 109171. IF: 11.0; ESI Highly Cited.
7. L. Chen, **Q. Li**, C. Shen et al. “Adversarial domain-invariant generalization: a generic domain-regressive framework for bearing fault diagnosis under unseen conditions.” *IEEE Transactions on Industrial Informatics*, 2021/2022. IF: 9.9; ESI Highly Cited.
8. **Q. Li**, C. Shen, L. Chen et al. “Knowledge mapping-based adversarial domain adaptation: A novel fault diagnosis method with high generalizability under variable working conditions.” *Mechanical Systems and Signal Processing*, 2021, 147. IF: 8.9; ESI Highly Cited.
9. **Q. Li**, X. Zhang, W. Hu et al. “Autonomous signal-processing large language model agents for mechanical vibration signals.” *Journal of Vibration Engineering*, 2025.
10. **Q. Li**, L. Chen, C. Shen et al. “Enhanced generative adversarial networks for fault diagnosis of rotating machinery with imbalanced data.” *Measurement Science and Technology*.

RESEARCH OUTPUTS, SERVICE, AND LEADERSHIP

- Led and contributed to open-source repositories for trustworthy diagnosis systems, PHM benchmarks, and industrial vibration-signal analysis in the large-model era.
- Reviewer for *Information Fusion*, *IEEE Transactions on Industrial Informatics*, *Mechanical Systems and Signal Processing*, *Advanced Engineering Informatics*, *IEEE Transactions on Industrial Electronics*, *IEEE Transactions on Instrumentation and Measurement*, *Neural Networks*, and *Engineering Applications of Artificial Intelligence*.
- Served as PHMAP 2025 session chair.
- New Student Representative at the Graduate Opening Ceremony; volunteer at the IFToMM International Conference on Rotor Dynamics; invited speaker at the Doctoral Forum of the Department of Mechanical Engineering and the Beijing AI PhD Student Forum.

ENGINEERING APPLICATIONS

- Aircraft vibration and noise reduction: contributed to sensor layout optimization, vibration monitoring, signal-processing assessment, and correlation-based evaluation for large-space aircraft structures.
- Hydropower intelligent operation and maintenance: contributed to modeling, simulation, fault-tree analysis, and trustworthy online monitoring for industrial cyber-physical systems.
- Industrial foundation models: developed physics-informed data generation, fine-tuning, evaluation, and interaction workflows for domain-specialized large models.